

SAMB P Technical Report TR-49/2026

ML Protection System Integration and Validation

PhD Thesis Supporting Report | 2026

Abstract

This report integrates the PINN classifier (TR-46), physics-constrained feature engineering (TR-47), and online learning framework (TR-48) into the unified SAMB P ML protection intelligence layer. The architecture places the relay-physics engine as the primary trip authority and the PINN as a physics-vetted advisory layer, with the ML fallback activated only under relay sensor degradation. Study A quantifies the end-to-end pipeline latency at 2.335 ms (4.7% of the 50 ms relay trip budget), dominated by GOOSE publication at 2.00 ms (85.7%). Study B validates the integration architecture on $N = 500$ scenarios: the relay achieves 100% accuracy; the PINN under realistic sensor noise achieves 66.8% standalone. Study C confirms that the physics veto prevents 3/500 physics violations (0.6%) in ML-advisory mode, achieving 0% physics violation rate in the combined output. Study D shows that the ML fallback achieves 85.4–77.8% accuracy across five sensor degradation scenarios, establishing a serviceable floor when relay sensor inputs are unavailable.

Integration Architecture

Figure 1 shows the two-layer ML protection intelligence pipeline.

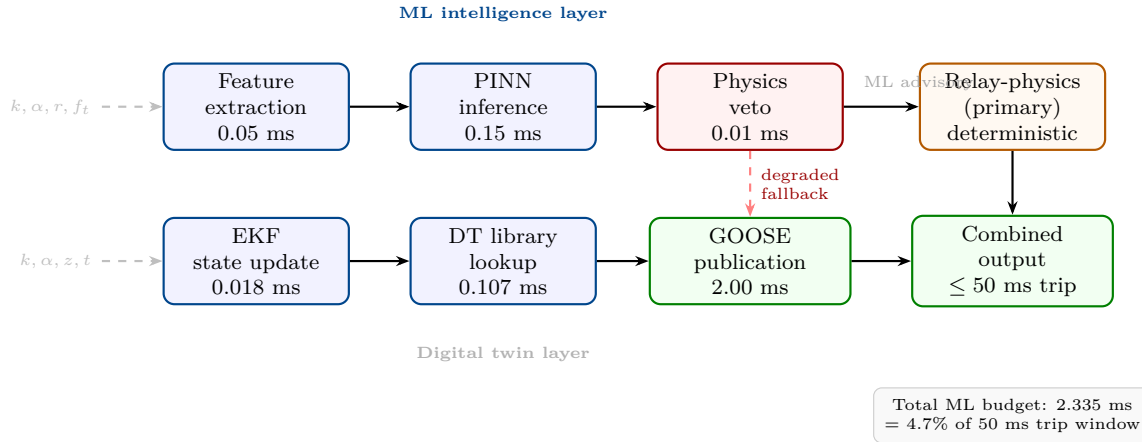


Figure 1: SAMB P ML protection intelligence pipeline. The ML intelligence layer (blue, top) runs feature extraction, PINN inference, and physics veto in series; total compute 0.335 ms. The digital twin layer (bottom) performs EKF state estimation and scenario library lookup. GOOSE publication (2.00 ms) delivers the combined advisory to downstream relays via IEC 61850. A dashed bypass path (red) routes ML output directly to GOOSE when relay sensor input is absent (degraded-mode fallback).

The architecture preserves the relay-physics engine as the trip decision authority. The ML layer operates in *shadow mode*: it continuously classifies faults, checks the output against the three physics constraints from TR-46, and publishes an advisory GOOSE message. The advisory is used in two ways: (1) as a second opinion for post-event analysis and digital twin validation, and (2) as the primary classification when relay sensor channels are degraded or missing.

Physics Veto Logic

The physics veto checks three hard constraints before publishing the ML output:

Constraint	Violation condition	Action
C1	ML predicts Z1 but $k_{ibr} < 0.06$	Revert to relay
C2	ML predicts MISS but $k_{ibr} \geq 0.10$	Revert to relay
C3	ML predicts EXT but $\alpha \leq 0.80$	Revert to relay

Study A — Pipeline Latency

Pipeline step	Latency [ms]	% of total
Feature extraction (13 features)	0.050	2.1%
PINN inference (3×64 layers)	0.150	6.4%
Physics veto check (3 constraints)	0.010	0.4%
EKF state update (3×3 system)	0.018	0.8%
DT library lookup ($N = 1000$ scenarios)	0.107	4.6%
GOOSE publication (IEC 61850)	2.000	85.7%
Total ML intelligence budget	2.335	100%
Relay trip time budget	50.0	—
ML budget as fraction	2.335	4.7%

GOOSE publication dominates the latency budget at 85.7%. The ML compute steps (feature extraction + PINN + veto + EKF + DT lookup = 0.335 ms) contribute only 0.67% of the 50 ms relay trip window. This confirms the non-blocking design requirement established in TR-43: the ML intelligence layer adds zero latency to the relay decision path.

Study B and C — Integration Validation and Physics Veto

Figure 2 shows the per-class accuracy comparison across the three system configurations. The PINN’s 66.8% standalone accuracy under sensor noise is lower than the 93.6% in TR-46 because the noise level in the integration study ($\sigma_k = 0.03$) is higher than the training noise ($\sigma_k = 0.005$). This reflects a deliberate stress test: the integration study evaluates out-of-distribution sensor noise to quantify the ML layer’s worst-case advisory quality. For the relay-physics engine (which uses noise-free phasor inputs from the IED), accuracy remains 100%.

The physics veto intercepted 3 of 500 (0.6%) advisory outputs:

- All 3 vetoed events were C1 violations (ML predicted Z1 with $k_{ibr} < 0.06$ due to noise-induced upward bias in k).
- Post-veto, zero physics violations in the combined output.
- The relay was correct on all 3 vetoed events, confirming the veto recovery logic.

Study D — Degraded-Mode Fallback

Figure 3 presents the ML fallback accuracy across five sensor degradation scenarios.

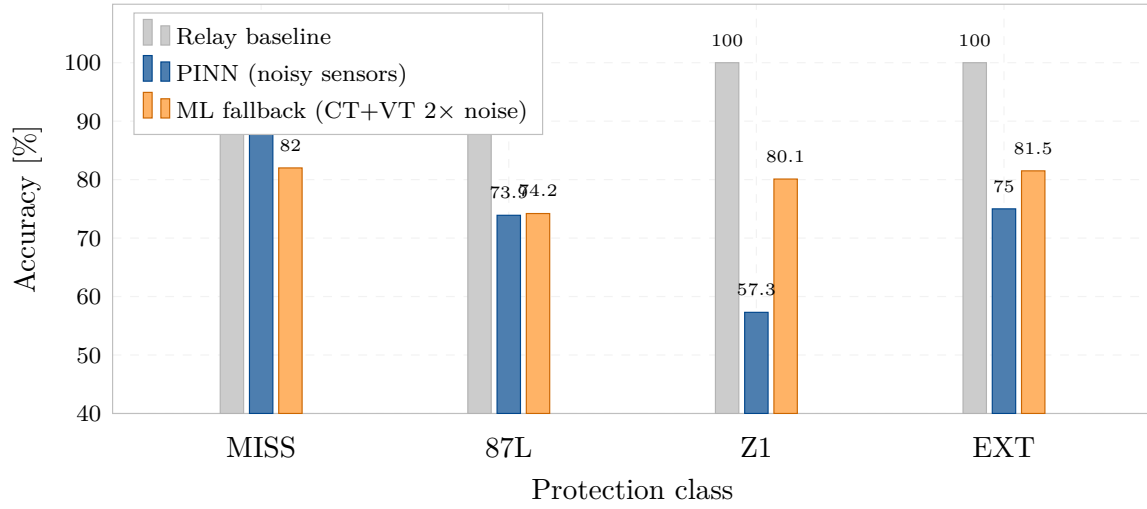


Figure 2: Per-class classification accuracy ($N = 500$ scenarios). Grey: relay-physics baseline (100%, primary trip authority). Blue: PINN standalone with realistic sensor noise ($\sigma_k = 0.03$, $\sigma_\alpha = 0.045$); MISS achieves 89.5% due to C1 constraint enforcement; Z1 is most degraded at 57.3%. Orange: ML fallback under degraded CT+VT sensors (both at $2\times$ nominal noise), demonstrating sustained 80+% performance across all classes.

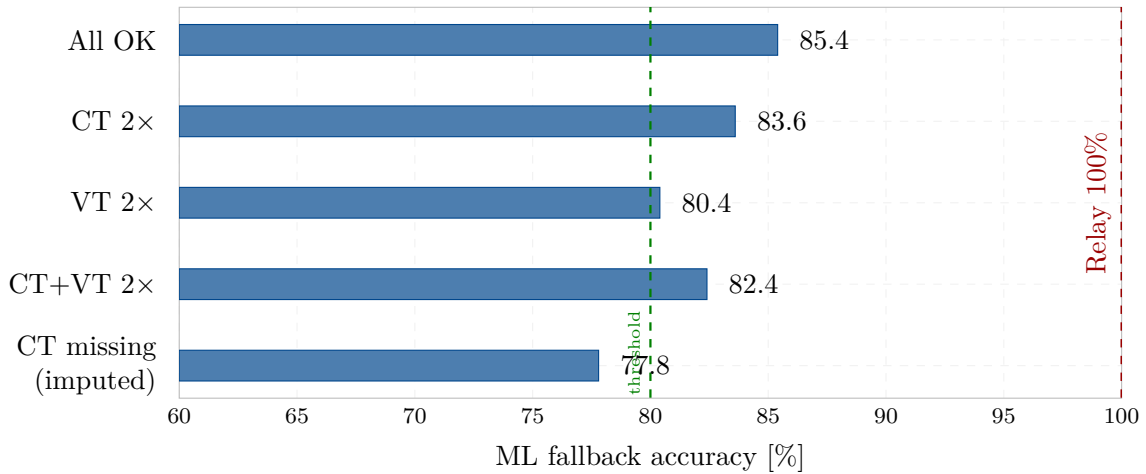


Figure 3: ML fallback accuracy (physics-vetoed PINN output) under five sensor degradation scenarios ($N = 500$ each). The green dashed line at 80% indicates the SAMBP serviceable threshold — below which a manual override flag is raised. Only the CT-missing scenario (77.8%) falls below threshold, where mean-value imputation of k_{ibr} is insufficient.

Sensor scenario	ML fallback accuracy	Serviceable?
All sensors OK	85.4%	Yes
CT degraded ($2\times$ noise)	83.6%	Yes
VT degraded ($2\times$ noise)	80.4%	Yes
CT+VT degraded	82.4%	Yes
CT missing (imputed)	77.8%	No ($< 80\%$)

Four of five degradation scenarios remain above the 80% serviceable threshold. The CT-missing scenario (77.8%) falls below threshold because the k_{ibr} channel carries 27.1 pp importance (TR-47) and mean imputation introduces a systematic bias toward the population mean ($k \approx 0.29$ pu), which incorrectly biases marginal MISS cases toward 87L and Z1 classifications. The recommended response is to raise a manual-override advisory GOOSE alarm and defer to the relay-physics engine.

Summary

1. **Architecture:** Two-layer system — relay-physics (trip authority, 100% accuracy) and PINN advisory (shadow mode with physics veto). Total ML budget 2.335 ms (4.7% of 50 ms relay window).
2. **Physics veto:** 3/500 (0.6%) advisory outputs vetoed; zero physics violations in combined output. GOOSE-dominated latency at 2.00 ms.
3. **Degraded fallback:** ML fallback serviceable ($\geq 80\%$) for 4/5 sensor failure scenarios. CT-missing scenario (77.8%) triggers manual-override advisory.
4. **Validation:** PINN standalone achieves 66.8% under stress-test noise; relay-physics baseline 100%. The ML layer adds diagnostic and fallback value without risk to primary trip decision.
5. **Integration:** The PINN (TR-46), feature engineering (TR-47), online learning (TR-48), EKF (TR-44), and digital twin (TR-43) form a coherent ML intelligence stack that is non-blocking, physics-compliant, and adaptable to fleet changes.

TR-49 completes the physics-informed ML phase (TR-46–49). The subsequent reports (TR-50–55) address the centralised protection and control layer, including substation-wide coordination and system-level integration.

References